Lakes and stations on map

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```
setwd("~/Documents/School/2025/dsp/ice-loss/")
DATADIR <- "./client/Data/"

library(tidyverse)
library(FactoMineR)
library(factoextra)
library(GGally)
library(ggplot2)
library(dplyr)
library(sf)
library(rnaturalearth)
library(rnaturalearthdata)
```

Pairwise plots of lake params

```
ltbl_ice <- read_rds(paste0(DATADIR, "ltbl_ice.rds"))

(lake_num <- ltbl_ice %>%
  select(-one_of(c('lake_id', 'station_id', 'country', 'subset', 'lake_name'))))[1:6, ]
```

```
##   lat_wgs84 lon_wgs84 altitude_m  area_ha depth_mean_m depth_max_m
## 1  66.85586  23.50141      183.8    33.36           NA          NA
## 2  63.73562  13.01327      400.9   780.83          9.7        51.0
## 3  66.05579  17.54014      420.9 24714.97           NA          NA
## 4  65.51011  15.62401      439.0  1561.67          8.9        18.5
## 5  63.69644  14.14094      472.9  1188.52           NA          NA
## 6  67.51683  18.17547      446.4 25830.46           NA        92.0
##   cent_lat_wgs84 cent_lon_wgs84
## 1       66.85303       23.50417
## 2       63.75034       12.94013
## 3       65.98060       17.72491
## 4       65.54448       15.56062
## 5       63.73229       14.10157
## 6       67.66406       17.57993
```

```
ltbl_ice %>%
  summarise_at(vars(lake_id, station_id, country, subset, lake_name), n_distinct)
```

```
##   lake_id station_id country subset lake_name
## 1    1197       1279       5     20      1235
```

```
ltbl_ice %>%
  group_by(country) %>%
```

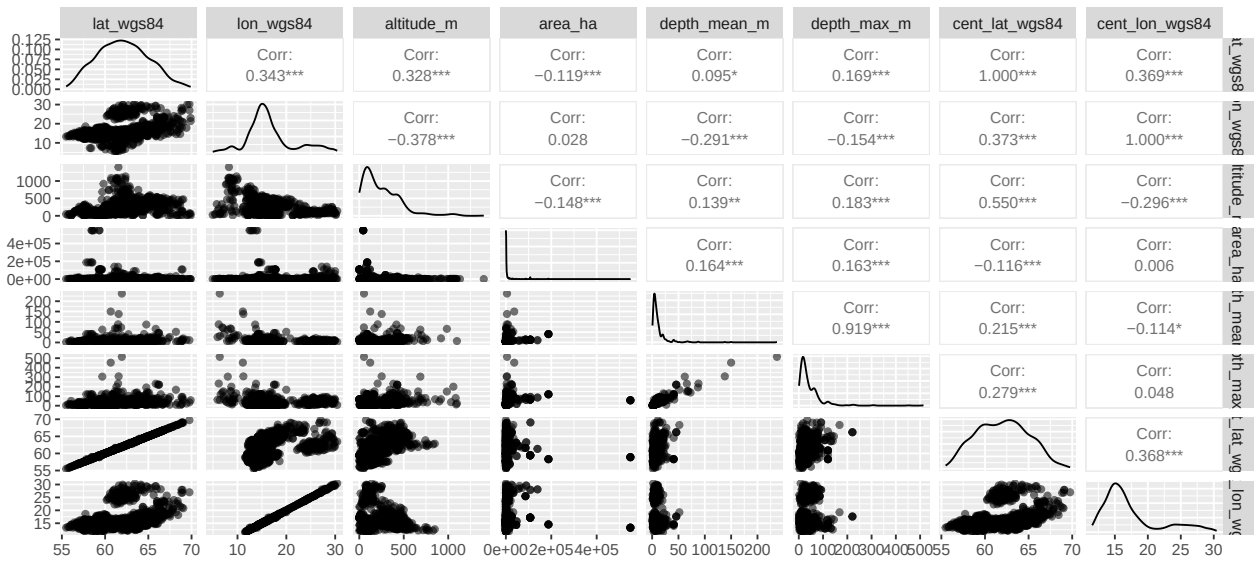
```r
  summarise(count = n()) %>%
  arrange(desc(count))
```

```
## # A tibble: 5 x 2
##   country count
##   <chr>   <int>
## 1 SE        679
## 2 CA        260
## 3 FI        121
## 4 US        117
## 5 NO        102
```

```r
ltbl_ice %>%
  group_by(subset) %>%
  summarise(count = n()) %>%
  arrange(desc(count))
```

```
## # A tibble: 20 x 2
##    subset  count
##    <chr>   <int>
##  1 <NA>      902
##  2 AK        107
##  3 ON         74
##  4 BC         31
##  5 IISD       30
##  6 QC         29
##  7 MB         24
##  8 NT         19
##  9 AB         10
## 10 NT-LTER    10
## 11 SK         10
## 12 NF          8
## 13 YT          8
## 14 NS          5
## 15 NB          4
## 16 NV          4
## 17 MN          1
## 18 NWT         1
## 19 QB          1
## 20 YK          1
```

```r
# limit to Fennoscandia
lake_fs_num <- lake_num %>%
  filter(lon_wgs84 > 0)

# ignore missing values for now
ggpairs(lake_fs_num, mapping = aes(alpha = 0.3),
             lower = list(combo = wrap("facethist", bins = 20)),
             upper = list(continuous = wrap("cor", size = 3)))
```



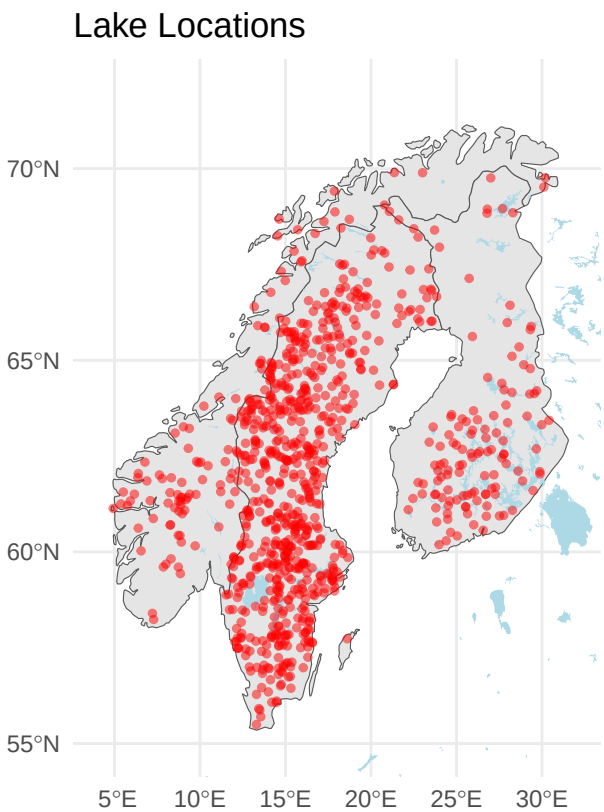
Lakes on map

Get map data.

```
fs_maps <- ne_countries(scale = "medium", returnclass = "sf") %>%
  filter(name %in% c("Sweden", "Finland", "Norway"))

lakes <- ne_load(scale = 10, type = "lakes", destdir = "maps")
```

Lakes and station (?) coords.

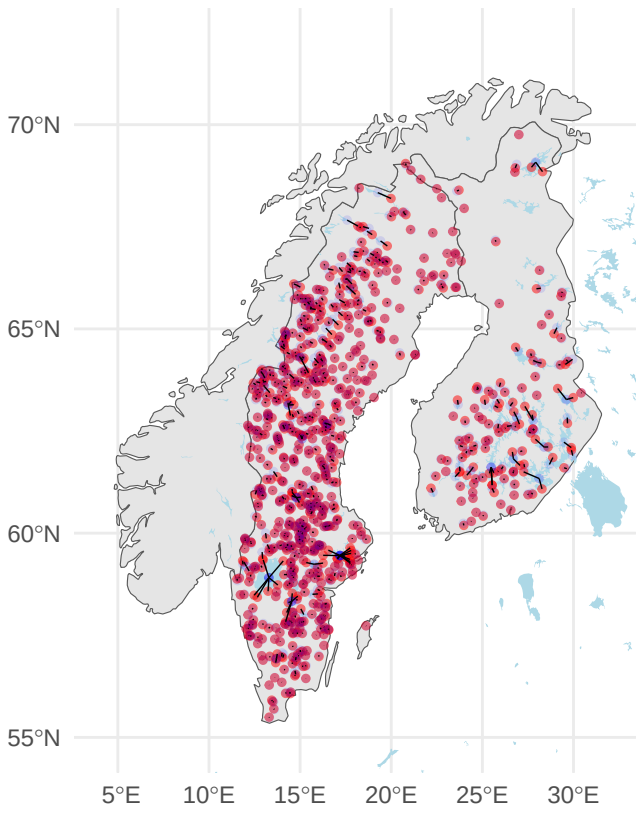
```
ggplot() +
  geom_sf(data = fs_maps) +
  geom_sf(data = lakes, fill = "lightblue", color = NA) +
  geom_point(data = lake_fs_num, aes(x = lon_wgs84, y = lat_wgs84),
             color = "red", size = 1, alpha=0.5) +
  coord_sf(xlim = c(4, 32), ylim = c(55, 72)) +
  theme_minimal() +
  labs(title = "Lake Locations") +
  xlab("") +
  ylab("")
```




```
# ggsave("images/lake-locations-with-lakes-fs.pdf", plot = g, bg = "transparent")

# figure margins: https://stackoverflow.com/questions/67687029/
```

Lakes, station (?) coords and lake cent coords.

```
# plot shifs in coordinates
# omit nan rows

lake_fs_num_shift <- lake_fs_num %>%
  filter(if_all(c(lon_wgs84, lat_wgs84, cent_lon_wgs84, cent_lat_wgs84), ~!is.na(.)))


ggplot() +
  geom_sf(data = fs_maps) +
  geom_sf(data = lakes, fill = "lightblue", color = NA) +
  geom_point(data = lake_fs_num_shift, aes(x = lon_wgs84, y = lat_wgs84),
             color = "red", size = 1, alpha=0.5) +
  geom_point(data = lake_fs_num_shift, aes(x = cent_lon_wgs84, y = cent_lat_wgs84),
             color = "blue", size = 1, alpha=0.1) +
  geom_segment(data = lake_fs_num_shift,
               aes(x = lon_wgs84, y = lat_wgs84, xend = cent_lon_wgs84, yend = cent_lat_wgs84),
               color = "black", size = 0.25) +
  coord_sf(xlim = c(4, 32), ylim = c(55, 72)) +
  theme_minimal() +
  xlab("") +
  ylab("")
```


70°N
65°N
60°N
55°N
5°E
10°E
15°E
20°E
25°E
30°E